

TAKE IT FURTHER

Instead of using paper-mache on a cardboard base, you could use mud to build up a volcano cone on a wooden base. In this version, just leave a hole in the top of the soil cone and press a plastic cup into it. This makes the crater where you will mix up the lava. If you're allowed to use lots of vinegar and baking soda, stir up a really large cupful of ingredients and create a truly gigantic eruption. Or you could see what happens if you replace baking soda with cola, which contains phosphoric acid.



When the flow starts to slow, keep the fun going by adding more baking soda and vinegar.

Red food coloring makes your foamy mixture actually look like lava.

The lava is easy to wipe up when the eruption is finally over.

HOW IT WORKS

Mixing baking soda (a base) with vinegar (which contains acetic acid) causes a rapid chemical reaction that produces lots of carbon dioxide gas. Tiny bubbles of this gas become trapped in the dishwashing liquid in your lava mixture. This creates a foam that takes up much more space than the liquid ingredients—so it all comes frothing out of the mouth of the volcano and trickles down the sides. Real lava has tiny bubbles of carbon dioxide in it, too. When it cools and sets hard, the bubbles are trapped.

REAL WORLD SCIENCE TUNGARAHUA VOLCANO



Your volcano is the shape of what is known as a cinder cone volcano. Tungurahua, shown in the picture, is an active volcano of this type in Ecuador, South America. When it erupts, lava and ash run down and solidify, forming another layer of rock to build up the growing cone.

INSIDE A VOLCANO

There are several types of volcano, but deep inside each one there is a large pool of molten rock, called the magma chamber. When underground pressures increase, the magma is pushed into a central tube and out through the crater as lava.

